In LLM Reasoning, there is Irrationality on top of Value Misalignment

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Abstract

Significant progress has been made in aligning LLMs with target value functions. We argue that, even when an LLM has been well aligned in (post-)training, it may still fail to maximise the aligned value in reasoning. We mathematically formalise this gap as rational value risk: the utility discrepancy between a model's deployed reasoning strategy and its rational counterpart whose responses maximise utility in the steepest direction. The estimation error of rational value risk is further decomposed into three components from bounded prompts, bounded responses, and imperfect verifiers. Extensive experiments are conducted, covering models Llama-3.1, Qwen-2.5, Tülu-3 families (7B-72B), GPT-5.2, GPT-5.5, and DeepSeek-V4, and benchmarks UltraFeedback, AlpacaEval, GSM8K, MATH, HumanEval, and MathArena. The results validate that (1) rational value risk is widespread; (2) value alignment can reduce, but cannot avoid, it; (3) self-consistency can improve rationality; and (4) a longer chain of thought improves rationality but with diminishing returns. The code is at `https://github.com/EVIEHub/LLM-Rationality`.

1 Introduction

Significant progress has been made in aligning large language models (LLMs) with target value functions. Through value alignment methods such as supervised fine-tuning (SFT) (Ouyang et al., 2022), reinforcement learning from human feedback (RLHF) (Christiano et al., 2017; Stiennon et al., 2020; Ouyang et al., 2022), direct preference optimisation (DPO) (Rafailov et al., 2023), constitutional training (Bai et al., 2022), and reinforcement learning with verifiable rewards (RLVR) (Shao et al., 2024), modern LLMs have become increasingly capable of producing outputs that better reflect human preferences, task-specific objectives, and safety constraints. These advances have substantially improved the helpfulness, reliability, and task performance of LLMs across open-ended dialogue (Thoppilan et al., 2022), mathematical reasoning (Shao et al., 2024), code generation (Chen et al., 2021), amongst others.

This paper argues that value alignment alone does not guarantee ideal reasoning at inference:

> *Even if an LLM has been well aligned in training, it may still fail to maximise the aligned value in reasoning.*

We mathematically formalise this phenomenon by *rational value risk*, defined as the utility discrepancy between a deployed reasoning strategy and its rational counterpart. Here, we define *rational reasoning* as the strategy whose responses maximise utility in the steepest direction under the given value function. This definition separates two sources of failure that are often conflated in LLM evaluation: (1) value misalignment: the learned value function itself is misaligned, and (2) irrational reasoning: the model fails to produce the responses that maximise the aligned value.

The rational value risk is not directly accessible. We define a compute-bounded estimator of expected utility using finite response samples and verifier evaluations, and use it to estimate rational value risk. The estimation error of this estimated rational value risk is decomposed into three components: (1) a prompt sampling error, which captures the statistical error from evaluating on finitely many prompts, (2) a reasoning sampling error, due to bounded sampling of the deployed and rational reasoning strategies, and (3) verification error, arising when the evaluation signal is uncertain or imperfect.

We conduct extensive experiments across both open-ended conversational tasks and verifiable reasoning tasks. The evaluation covers the Llama-3.1 (Grattafiori et al., 2024), Qwen-2.5 (Qwen et al., 2025), and Tülu-3 model (Lambert et al., 2025) families from 7B to 72B parameters, as well as proprietary models including GPT-5.2 (OpenAI, 2025), GPT-5.5 (OpenAI, 2026), DeepSeek-V4-Pro (DeepSeek-AI et al., 2026), and DeepSeek-V4-Flash (DeepSeek-AI et al., 2026). The benchmarks include UltraFeedback (Cui et al., 2024) and AlpacaEval (Dubois et al., 2024) for conversational preference evaluation, GSM8K (Cobbe et al., 2021) and MATH (Hendrycks et al., 2021) for mathematical reasoning, HumanEval (Chen et al., 2021) for code generation, and MathArena (Balunović et al., 2025) for challenging deployment-style mathematical reasoning.

The empirical results validate four hypotheses:

H1: Rational value risk is widespread. Across eight models of different scales and six benchmarks, LLM reasoning consistently deviates from the rational counterpart, resulting in utility loss.

H2: Value alignment methods can reduce, but cannot avoid, rational value risk. Rationality is not fully solved by value alignment alone.

H3: Rationality benefits from consistency across reasoning paths, and is insensitive to sampling diversity (or temperature).

H4: A longer chain of thought can improve rationality, but its benefits diminish beyond a certain reasoning budget.

Together, these findings are the first in the literature to formally define and empirically measure rational value risk in LLM reasoning, separating inference-time irrationality from value misalignment, to the best of our knowledge.

2 Related works

Rationality in machine learning. Valiant (1995) offers a philosophical account of rationality under a PAC-style criterion. Abel (2019) defines bounded rationality in reinforcement learning and also shows that rational decision-making involves a trade-off between representational simplicity and predictive accuracy. From behavioural data, Evans et al. (2025) model bounded rationality through a Wasserstein constraint between the learned policy and a prior. Sunehag and Hutter (2015) derive decision-theoretic axioms for rational reinforcement-learning agents, although these axioms exclude many standard algorithms.

Qian et al. (2026) design tractable rationality measures and develop theory for reinforcement learning agents. The results, however, do not apply to LLM reasoning in deployment. This work addresses the gap through a new suite of rationality measures, theory, and extensive experiments, as part of our continuing efforts to understand the rationality of AI agents.

Rationality in LLMs. Cognitive-science evaluations test whether models exhibit human-like bounded rationality, heuristics, content effects, and cognitive biases (Binz and Schulz, 2023; Hagendorff et al., 2023; Macmillan-Scott and Musolesi, 2024; Yax et al., 2024; Lampinen et al., 2024; Coda-Forno et al., 2024; Malberg et al., 2025; Brady et al., 2025). Decision-theoretic and economic approaches instead ask whether model choices are consistent with utility maximisation, risk attitudes, revealed preferences, or preference axioms such as transitivity (Chen et al., 2023; Jia et al., 2024; Song et al., 2025; Liu et al., 2025). A third line audits or improves rational behaviour by enforcing belief consistency, logical preference consistency, debiasing, or rational thought prompting (Kassner et al., 2023; Echterhoff et al., 2024; Koo et al., 2024; Gou et al., 2024). Recent surveys and benchmarks consolidate these views into broader notions of rational agents with consistency, grounding, preference orderability, and evidence-aligned decision making (Jiang et al., 2025; Zhou et al., 2025).

Although prior work has shown that LLMs exhibit rationality failures such as inconsistency, cognitive biases, and violations of decision-theoretic principles, it has not explicitly characterised these failures as utility loss that persists after value alignment. In contrast, our work mathematically formalises rationality as an inference-time utility gap between a deployed reasoning strategy and its rational counterpart, thereby separating irrational reasoning from value misalignment.

Evaluation of LLM reasoning. The evaluation of LLM reasoning commonly relies on sampling-based performance metrics. Pass@1 reports the success rate of a single sampled response, while pass@k reports the probability of obtaining at least one correct response within k samples (Chen et al., 2021). Oracle best-of-k similarly evaluates the best performance among a finite set of sampled candidates under oracle selection (Sunkaraneni et al., 2026). These metrics and their variants quantify the probability or coverage of successful responses from a finite set of samples, and typically rely on binary success signals for tasks with deterministic or verifiable outcomes.

3 Preliminaries

Let $\mathcal{X}$ denote the space of input prompts and $\mathcal{V}$ a finite vocabulary. The reasoning space is $\mathcal{Z} = \bigcup_{T\geq 1} \mathcal{V}^T$, where T denotes the length of a reasoning path $z = (z_1, \ldots, z_T)$.

In reasoning, a frozen language model π_θ parameterised by $\theta \in \Theta$, where Θ is the parameter space, generates a reasoning path $z \in \mathcal{Z}$ by sequentially sampling tokens according to a conditional probability on $x \in \mathcal{X}$:

$$\pi_\theta(z \mid x) = \prod_{t=1}^{T} \pi_\theta\left(z_t \mid x, z_{<t}\right).$$

Given a reasoning problems $D = \{\mathbf{x}_i\}_{i=1}^M$, where $\mathbf{x}_i = (x_i, y_i^+)$ of M input questions $x_i \in \mathcal{X}$ and preferred (or verifiable) answer $y_i^+ \in \mathcal{Y}$, we define a reasoning strategy

$$d_\theta(\cdot|x_i) \triangleq d(\pi_\theta(\cdot|x_i)),$$

including temperature sampling, self-consistency, and varying context length. For each input question x_i, a frozen language model generates a reasoning path z_i or answer $y_i = g(z_i)$ from an extraction function $g : \mathcal{Z} \to \mathcal{Y}$ by reasoning strategy $d_\theta(\cdot|x_i)$.

Let a $\mathcal{O}$ be the outcome space, and a verifier is modelled as an outcome distribution

$$P : \mathcal{X} \times \mathcal{Y} \times \mathcal{Y} \to \Delta(\mathcal{O})$$

evaluates the generated answer and returns an outcome $o_i \sim P(\cdot|\mathbf{x}_i, y_i)$. This formulation covers preference-based tasks, where outcomes may be stochastic due to variation across annotators, judges, or repeated evaluations. In verifiable reasoning tasks, such as mathematical reasoning and code generation, the outcome distribution degenerates to a Dirac distribution as

$$\delta_{f(\mathbf{x}_i, y_i)}(o_i) : P(o_i = f(\mathbf{x}_i, y_i)|\mathbf{x}_i, y_i) = 1,$$

where

$$f : \mathcal{X} \times \mathcal{Y} \times \mathcal{Y} \to \mathcal{O}$$

is a deterministic verifier. Let $U : \mathcal{O} \to \mathbb{R}$ be a utility function defined on the outcome space, assigning a utility value $U(o_i) \in [0, 1]$ to each evaluation outcome $o_i \in \mathcal{O}$.

4 Rationality of LLM reasoning

This section defines the rationality measurement for LLM reasoning.

4.1 Rationality measures

We first define (perfectly) rational reasoning.

Definition 1 (rational reasoning)**.** *A reasoning path z° with an answer $y^\circ = g(z^\circ)$ is called (perfectly) rational, if and only if its reasoning strategy maximises the expected utility function $U : \mathcal{O} \to \mathbb{R}$ over the outcome distribution $P(\cdot|\mathbf{x}, y)$:*

$$y^\circ \sim d_{\theta^\circ}(\cdot|x),$$
$$\theta^\circ \in \arg\max_{\theta\in\Theta} \mathbb{E}_{\mathbf{x}\sim\rho, y\sim d_\theta, o\sim P(\cdot|\mathbf{x}, y)} U(o).$$

LLM reasoning is not always rational. We define a rational value risk to quantify the discrepancy of expected utility between an LLM reasoning strategy and its rational counterpart.

Definition 2 (rational value risk)**.** *For any reasoning question $\mathbf{x} = (x, y^+)$ drawn from a distribution ρ, let y° denote an answer of rational reasoning under $d_\theta^\circ(\cdot \mid x)$, and y be an answer drawn from a reasoning strategy $d_\theta(\cdot \mid x)$. We define the rational value risk $\mathcal{R}(d_\theta)$ of d_θ under a utility function U as:*

$$\mathbb{E}_{\substack{\mathbf{x}\sim\rho,\, y^\circ\sim d_{\theta^\circ}(\cdot|\mathbf{x}),\\ o\sim P(\cdot|\mathbf{x}, y^\circ)}} [U(o)] - \mathbb{E}_{\substack{\mathbf{x}\sim\rho,\, y\sim d_\theta(\cdot|\mathbf{x}),\\ o\sim P(\cdot|\mathbf{x}, y)}} [U(o)].$$

The rational value risk is not directly accessible because it is a population term. To estimate it, we first define a compute-bounded estimated utility given a compute budget of K samplings.

Definition 3 (compute-bounded estimated utility)**.** *Let d_θ denote the reasoning strategy of a language model π_θ. For any reasoning problem $\mathbf{x}$, let*

$$y_1, \ldots, y_k \overset{iid}{\sim} d_\theta(\cdot \mid x),$$

where each $y_k = g(z_k)$ is the extracted answer from an independently and identically distributed (iid) sampled reasoning path z_k. For each sampled answer y_k, suppose we obtain L independently and identically distributed evaluation outcomes,

$$o_{k,1}, \ldots, o_{k,L} \overset{iid}{\sim} P(\cdot \mid \mathbf{x}, y_k).$$

Given a compute budget of K samplings, we define a compute-bounded estimated utility of $\{y_i\}_{i=1}^K$ as:

$$\frac{1}{K}\sum_{k=1}^{K} \widehat{U}_L(\mathbf{x}, y_k) = \frac{1}{L}\sum_{k=1}^{K}\sum_{l=1}^{L} U(o_{k,l}).$$

A Monte Carlo estimator is defined below to estimate the rational value risk.

Definition 4 (estimated rational value risk)**.** *Given a set of reasoning problems* $\{\mathbf{x}_i\}_{i=1}^M$*, let*

$$\{y_{i,k}\}_{k=1}^K \overset{iid}{\sim} d_\theta(\cdot \mid x_i)$$

denote K *sampled answers for each input* x_i*. Let*

$$\{y_{i,k}^\circ\}_{k=1}^K \overset{iid}{\sim} d_{\theta^\circ}(\cdot|x_i)$$

be the rational reasoning answers, and $\widehat{U}_{K,L}$ *be the compute-bounded estimated utility among the* K *sampled candidates in Definition 3. The estimated rational value risk* $\widehat{\mathcal{R}}_{M,K,L}(d_\theta)$ *under a utility function* U *is defined as*

$$\frac{1}{MK}\sum_{i=1}^M\sum_{k=1}^K\left[\widehat{U}_L(\mathbf{x}_i, y_{i,k}^\circ) - \widehat{U}_L(\mathbf{x}_i, y_{i,k})\right].$$

4.2 Theoretical guarantee on estimation

This section studies the estimation error in empirically computing the rational value risk. Detailed proofs are given in Appendix B.

The estimation error can be decomposed into three components: prompt sampling error, reasoning sampling error, and verification error.

Lemma 1 (estimation error decomposition)**.** *The estimation error decomposes as:*

$$\begin{aligned}
&\mathcal{R}(d_\theta) - \widehat{\mathcal{R}}_{M,K,L}(d_\theta)\\
&= \underbrace{\mathcal{R}(d_\theta) - \mathcal{R}_M(d_\theta)}_{\text{(I) } \textit{prompt sampling error}} + \underbrace{\mathcal{R}_M(d_\theta) - \overline{\mathcal{R}}_{M,K}(d_\theta)}_{\text{(II) } \textit{reasoning sampling error}}\\
&\quad + \underbrace{\overline{\mathcal{R}}_{M,K}(d_\theta) - \widehat{\mathcal{R}}_{M,K,L}(d_\theta)}_{\text{(III) } \textit{verification error}}
\end{aligned}$$

The three terms are respectively bounded below.

Lemma 2 (prompt sampling error bound)**.** *Given* M *reasoning problems* $\{\mathbf{x}_i\}_{i=1}^M$ *for any* $\delta \in (0,1)$*, with probability at least* $1-\delta$*, the prompt sampling error can be bounded by,*

$$|\mathcal{R}(d_\theta) - \mathcal{R}_M(d_\theta)| \le \sqrt{\frac{2\log(2/\delta)}{M}}.$$

Lemma 3 (reasoning sampling error bound)**.** *For any* $\delta \in (0,1)$*, given the sampled reasoning problems and* K *sampled reasoning responses for each problem, with probability at least* $1-\delta$*, the reasoning sampling error can be bounded by,*

$$\left|\mathcal{R}_M(d_\theta) - \overline{\mathcal{R}}_{M,K}(d_\theta)\right| \le \sqrt{\frac{2\log(2/\delta)}{MK}}.$$

Lemma 4 (verification error bound)**.** *Assume that all evaluation outcomes are sampled independently, given the sampled reasoning problems and reasoning answers, with* L *verifier evaluations for each sampled response. Then, for any* $\delta \in (0,1)$*, with probability at least* $1-\delta$*, the verification error can be bounded by,*

$$\left|\overline{\mathcal{R}}_{M,K}(d_\theta) - \widehat{\mathcal{R}}_{M,K,L}(d_\theta)\right| \le \sqrt{\frac{2\log(2/\delta)}{MKL}}.$$

If the verifier is deterministic, this term is zero.

We thus obtain the estimation guarantee.

Theorem 1 (estimation error bound)**.** *Assume* $U(o) \in [0,1]$*. For any* $\delta \in (0,1)$*, with probability at least* $1-\delta$ *over the sampled reasoning problems, reasoning answers, and evaluation outcomes,*

$$\begin{aligned}
&\left|\mathcal{R}(d_\theta) - \widehat{\mathcal{R}}_{M,K,L}(d_\theta)\right|\\
&\le \sqrt{\frac{2\log(6/\delta)}{M}} + \sqrt{\frac{2\log(6/\delta)}{MK}} + \sqrt{\frac{2\log(6/\delta)}{MKL}}.
\end{aligned}$$

If the verifier is deterministic, the verification error vanishes.

Sample complexity. To make the three statistical error terms at most ϵ with high probability, it suffices to take

$$\begin{aligned}
M &= O\big(\epsilon^{-2}\log(1/\delta)\big),\\
K &= O\big(\epsilon^{-2}\log(1/\delta)/M\big),\\
L &= O\big(\epsilon^{-2}\log(1/\delta)/MK\big).
\end{aligned}$$

Thus, estimating rational value risk depends on the numbers of reasoning problems, sampled reasoning responses, and verifier evaluations. For verifiable reasoning tasks with a deterministic verifier, the dependence on verifier samples is eliminated.

Practical insights. Theorem 1 shows that the estimation error decreases as the reasoning problem number M, sampled reasoning response number K, and verifier evaluation number L increase. Among them, the number of reasoning problems M is the primary factor, as increasing M reduces all three error terms simultaneously. Increasing K further reduces reasoning and verification errors, while increasing L reduces only the verification error. Consequently, when M and K are sufficiently large, verification sampling contributes relatively little to the total estimation error. However, increasing L only improves estimation of the verifier's expected utility $\mathbb{E}_{o\sim P(\cdot|\mathbf{x},y)}[U(o)]$ and cannot remove bias in the verifier itself.

5 Experiments

Extensive experiments are conducted to verify four empirically verifiable hypotheses, which well establish that *there is irrationality on top of misalignment in LLM reasoning.*

5.1 Empirically verifiable hypotheses

H1: Rational value risk is widespread. LLM reasoning in deployment can deviate from rational reasoning, leading to a utility loss. This utility loss is expected to appear across different models and benchmarks, including both conversational and verifiable reasoning tasks.

H2: Value alignment can reduce, but cannot avoid, irrationality. Value alignment shifts the model distribution toward the target value function, but the deployed reasoning strategy can fail to consistently produce responses that maximise that function. Thus, rational value risk remains after value alignment.

H3: Rationality benefits from self-consistency, and is insensitive to sampling diversity. With the LLM parameters fixed, inference-time reasoning strategies can affect its rationality. We study two strategies with different effects: sampling temperature changes the randomness and diversity of sampled reasoning responses, while self-consistency aggregates multiple responses to determine the final answer.

H4: A longer chain of thought improves rationality with diminishing returns. Increasing the reasoning length can help the model reduce rational value risk, but this effect diminishes after a certain reasoning budget.

5.2 Implementation details

Setup. We evaluate the rationality of LLM reasoning on two task types: **(1) conversational tasks**: given a conversation dataset $D = \{\mathbf{x}_i\}_{i=1}^{M}$ ($\mathbf{x}_i = (x_i, y_i^+)$), a stochastic verifier compares the LLM's answer y with the human preferred answer y_i^+, and then returns an outcome $o_i \in \{\text{win}, \text{tie}, \text{lose}\}$, where $o_i \sim P(\cdot|\mathbf{x}_i, y_i)$. We then define the utility of each outcome according to the verifier's preference:

$$U(o_i) = \begin{cases} 1, & o_i = \text{win} \quad (y_i \succ y_i^+) \\ 0.5, & o_i = \text{tie} \quad (y_i \approx y_i^+) \\ 0, & o_i = \text{lose} \quad (y_i \prec y_i^+) \end{cases},$$

where $\succ$, $\prec$, and $\approx$ denote verifier preference, non-preference, and indifference, respectively.

The stochastic verifier can be from the LLM itself, reflecting its subjective preference in $P(\cdot|\mathbf{x}_i, y_i)$, or from larger-scale models as external verifiers. **(2) verifiable reasoning tasks:** we consider binary utility functions defined by answer correctness: $U(f(\mathbf{x}_i, y_i)) = \mathbb{1}\left[y_i = y_i^+\right]$.

Datasets and benchmarks. For conversational tasks, we measure the rational value risk on two benchmarks: UltraFeedback (Cui et al., 2024) and AlpacaEval (Dubois et al., 2024). For verifiable reasoning, we use three widely used benchmarks in LLM development and evaluation: GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021), and HumanEval (Chen et al., 2021). In addition, we adopt one reasoning benchmark as a deployment dataset: MathArena, including AIME 2025 and BRUMO 2025 (Balunović et al., 2025). *Its release dates are later than or close to those of the evaluated models*, while it is more difficult than other benchmarks, inducing a distribution shift between the development inputs, $\mathbf{x} \sim \rho$, and (unseen or harder) deployment inputs, $\mathbf{x} \sim \rho^\dagger$. Detailed descriptions are in Appendix E.

Models. Experiments for H1-H4 use open-weight models, Qwen2.5-7B-Instruct (Qwen et al., 2025), Llama-3.1-8B-Instruct (Grattafiori et al., 2024), Llama-3.1-Tülu-3-8B-(SFT, DPO, RLVR) (Lambert et al., 2025), Qwen2.5-72B-Instruct (Qwen et al., 2025), and Llama-3.1-Tülu-3-70B-(SFT, DPO, RLVR) (Lambert et al., 2025), and proprietary models, GPT-5.2 (OpenAI, 2025), GPT-5.5 (OpenAI, 2026), DeepSeek-V4-Pro (DeepSeek-AI et al., 2026), and DeepSeek-V4-Flash APIs (DeepSeek-AI et al., 2026). In practice, the rational reasoning is not directly accessible. For conversational tasks, we assume that a stronger model is more likely to approximate rational reasoning and use DeepSeek-V4-Pro (1.6T parameters) (DeepSeek-AI et al., 2026), as a practical proxy. We use DeepSeek-V4-Flash (284B parameters) (DeepSeek-AI et al., 2026) as the external stochastic verifier for all evaluated models. For verifiable reasoning tasks, we set the expected utility of rational reasoning to 1, since ground-truth answers exist and rational reasoning achieves maximum utility by producing them.

Configuration. Unless otherwise specified, K=64 reasoning paths are sampled per prompt

Table 1: Rational value risk (RVR) across LLMs on conversational and development benchmarks. For conversational tasks, expected utility by rational reasoning (REU) is estimated using DeepSeek-V4-Pro, whereas for verifiable reasoning tasks, REU is set to 1.000. Compute budget K=64; values are reported as mean $\pm$ 95% bootstrap confidence interval. Bold indicates the smallest RVR per dataset. (Self.: self-as-verifier; Ext.: external verifier, DeepSeek-V4-Flash.)

		Llama-3.1-8B-Instruct			Qwen2.5-7B-Instruct			Tülu-3-8B-RLVR			Qwen2.5-72B-Instruct		
Task	Dataset	REU	DEU	RVR	REU	DEU	RVR	REU	DEU	RVR	REU	DEU	RVR
Conversation	UltraFeedback (self.)	0.643	0.524	0.119 $\pm$0.029	0.567	0.533	**0.034** $\pm$0.020	0.600	0.532	0.068 $\pm$0.024	0.782	0.712	0.069 $\pm$0.034
	UltraFeedback (ext.)	0.713	0.492	0.221 $\pm$0.026	0.713	0.504	0.208 $\pm$0.026	0.713	0.497	0.215 $\pm$0.027	0.713	0.590	**0.122** $\pm$0.030
	AlpacaEval (self.)	0.812	0.690	0.122 $\pm$0.020	0.793	0.660	0.133 $\pm$0.021	0.763	0.756	0.007 $\pm$0.020	0.970	0.972	**-0.002** $\pm$0.014
	AlpacaEval (ext.)	0.904	0.502	0.402 $\pm$0.017	0.904	0.543	0.361 $\pm$0.020	0.904	0.738	0.166 $\pm$0.019	0.904	0.824	**0.079** $\pm$0.017
Verifiable reasoning	GSM8K	1.000	0.780	0.220 $\pm$0.014	1.000	0.906	0.094 $\pm$0.012	1.000	0.861	0.139 $\pm$0.014	1.000	0.958	**0.042** $\pm$0.009
	MATH	1.000	0.470	0.530 $\pm$0.019	1.000	0.898	0.102 $\pm$0.013	1.000	0.658	0.342 $\pm$0.021	1.000	0.938	**0.062** $\pm$0.011
	HumanEval	1.000	0.431	0.569 $\pm$0.056	1.000	0.749	**0.251** $\pm$0.052	1.000	0.398	0.602 $\pm$0.047	1.000	0.730	0.270 $\pm$0.059
	MathArena	1.000	0.003	0.997 $\pm$0.002	1.000	0.087	0.913 $\pm$0.051	1.000	0.008	0.992 $\pm$0.007	1.000	0.115	**0.885** $\pm$0.063

Table 2: Rational value risk on MathArena across all evaluated models, including open-weight 7-8B and 70-72B models and proprietary models. As MathArena is a verifiable reasoning task, REU=1. Values are reported as mean $\pm$ 95% bootstrap confidence intervals.

Size	Model	Rational value risk
7-8B	Qwen2.5-7B	0.913 $\pm$0.051
	Tülu-3-8B-RLVR	0.992 $\pm$0.007
	Llama-3.1-8B	0.997 $\pm$0.002
70-72B	Qwen2.5-72B	0.885 $\pm$0.063
	Tülu-3-70B-RLVR	0.940 $\pm$0.042
APIs	DeepSeek-V4-Flash	0.414 $\pm$0.102
	GPT-5.2	0.510 $\pm$0.106
	GPT-5.5	0.431 $\pm$0.107

using temperature sampling with τ=1.0, without top-p or top-k truncation. We consider two verifier settings on UltraFeedback and AlpacaEval: (1) self-as-verifier with verifier budget L=5, where the evaluated model acts as its own verifier, and (2) external verifier using a larger-scale language model, such as DeepSeek-V4-Flash as the verifier with budget L=3. Estimator calibration and verifier details are presented in Appendix D and Appendix C.3, respectively.

Reproducibility and budget. Experiments are performed with vLLM 0.6.3 on six NVIDIA A800 80GB GPUs and APIs including GPT 5.2, GPT 5.5, DeepSeek-V4-Pro, and DeepSeek-V4-Flash. The experiments require approximately 76.6 GPU-hours. API cost is $583 in total, with GPT 5.2($117), GPT 5.5($117), DeepSeek-V4-Pro($77), and DeepSeek-V4-Flash($272). Additional implementation details are provided in Appendix C. The code is at `https://github.com/EVIEHub/LLM-Rationality`.

5.3 Experimental results

H1: Rational value risk is widespread. Table 1 shows the expected utility of rational reasoning (REU), expected utility of deployed reasoning (DEU), and their difference, rational value risk (RVR), across conversational, mathematical reasoning, and code generation benchmarks. We observe positive rational value risk across nearly all evaluated settings.

On conversational tasks, RVR is generally larger under external verification than under self-verification. For example, on UltraFeedback, externally verified RVR ranges from 0.122 for Qwen2.5-72B to 0.504 for Qwen2.5-7B, but self verification has smaller RVR values for all models, including 0.068 for Tülu-3-8B and 0.069 for Qwen2.5-72B. A similar pattern holds on AlpacaEval, where Qwen2.5-72B exhibits an RVR close to zero (-0.002) in the self-as-verifier setting. This pattern suggests a self-verification bias that models may prefer their own responses.

On verifiable reasoning tasks, RVR increases when models are evaluated on more challenging deployment benchmarks. For instance, Qwen2.5-72B exhibits RVRs of only 0.042 on GSM8K and 0.062 on MATH, but it rises markedly to 0.885 on MathArena. This indicates that model reasoning becomes less rational as task difficulty increases.

Table 2 further compares RVR across models of different scales on MathArena. The evaluated 7-8B models exhibit the highest RVR, ranging from 0.913 to 0.997. The 70-72B models show only slight reductions, with RVRs of 0.885 and 0.940. Proprietary models achieve the lowest RVR, ranging from 0.414 to 0.510, but still exhibit a large utility gap relative to rational reasoning on challenging tasks.

These results confirm that rational value risk is

Table 3: Rational value risk of Tülu-3-8B family across SFT, DPO, and RLVR stages. Bold indicates the smallest RVR per dataset. Values are reported as mean ± 95% bootstrap confidence interval. (Ext.: external verifier, DeepSeek-V4-Flash).

Dataset		SFT	DPO	RLVR
UltraFeedback (external verifier)	REU	0.713	0.713	0.713
	DEU	0.037	0.493	0.497
	RVR	0.675 ±0.027	0.219 ±0.026	**0.215** ±0.027
AlpacaEval (external verifier)	REU	0.904	0.904	0.904
	DEU	0.020	0.728	0.738
	RVR	0.884 ±0.019	0.176 ±0.020	**0.166** ±0.019
GSM8K	REU	1.000	1.000	1.000
	DEU	0.588	0.858	0.861
	RVR	0.412 ±0.016	0.142 ±0.013	**0.139** ±0.014
MATH	REU	1.000	1.000	1.000
	DEU	0.276	0.639	0.658
	RVR	0.724 ±0.016	0.361 ±0.021	**0.342** ±0.021
HumanEval	REU	1.000	1.000	1.000
	DEU	0.148	0.250	0.398
	RVR	0.852 ±0.021	0.750 ±0.036	**0.602** ±0.047

widespread across both conversational and verifiable reasoning tasks. They also reveal three key factors associated with reasoning rationality, including task difficulty, model capability, and, for conversational tasks, the choice of verifier. In this context, rational value risk becomes a central bottleneck in inference-time reasoning.

H2: Value alignment reduces, but cannot avoid, irrationality. Table 3 reports REU, DEU, and RVR of the Tülu-3-8B family across SFT, DPO, and RLVR stages on conversational and verifiable reasoning benchmarks. We observe that post-training generally reduces rational value risk, with the largest improvements occurring from SFT to DPO. For example, RVR decreases from 0.675 to 0.219 on UltraFeedback, from 0.412 to 0.142 on GSM8K, and from 0.724 to 0.361 on MATH. The additional reduction from DPO to RLVR is modest on these benchmarks, although HumanEval shows a larger decrease, from 0.750 to 0.602. Despite these improvements, rational value risk remains after RLVR, particularly on MATH and HumanEval.

Table 4 reports rational value risk for the Tülu-3 8B and 70B models across the value-alignment pipeline on the more challenging MathArena benchmark. We observe a different pattern from the previous benchmarks, with value alignment showing limited reductions in RVR. For Tülu-3-8B,

Table 4: Rational value risk along the value-alignment pipeline of the Tülu-3 models on MathArena. Values are reported as mean ± 95% bootstrap confidence interval.

Model	Stage	Rational value risk
Tülu-3-8B	SFT	0.995 ±0.003
	DPO	**0.989** ±0.008
	RLVR	0.992 ±0.007
Tülu-3-70B	SFT	0.976 ±0.018
	DPO	**0.938** ±0.042
	RLVR	0.940 ±0.042

Table 5: Effects of sampling temperature $\tau = \{0.0, 0.7, 1.0\}$ on rational value risk of Tülu-3-8B-RLVR (Tülu-3-8B), Qwen2.5-7B-Instruct (Qwen2.5-7B), and Llama-3.1-8B-Instruct (Llama-3.1-8B). Values are reported as mean ± 95% bootstrap confidence interval.

Dataset	Model	τ=0.0	τ=0.7	τ=1.0
GSM8K	Llama-3.1-8B	0.154	0.170	0.220
	Qwen2.5-7B	0.082	0.088	0.094
	Tülu-3-8B	0.136	0.133	0.139
MATH	Llama-3.1-8B	0.344	0.392	0.530
	Qwen2.5-7B	0.089	0.093	0.102
	Tülu-3-8B	0.328	0.331	0.342
MathArena	Llama-3.1-8B	1.000	0.989	0.997
	Qwen2.5-7B	0.900	0.906	0.913
	Tülu-3-8B	1.000	0.989	0.992

RVR remains nearly unchanged across SFT, DPO, and RLVR (0.995, 0.989, and 0.992, respectively). Tülu-3-70B shows a larger reduction from 0.976 at SFT to 0.938 at DPO, but little additional change after RLVR (0.940). It shows that rationality improvement of Tülu-3-8B is not apparent across the alignment stages, suggesting that value alignment is less effective on challenging benchmarks at the 8B scale.

These results indicate that value alignment improves reasoning rationality, particularly through DPO, but rational value risk persists even after RLVR. It is also less effective at reducing rational value risk under increased task difficulty.

H3: Rationality benefits from self-consistency, but is insensitive to sampling diversity. Table 5 reports the effect of sampling temperature ($\tau \in \{0, 0.7, 1.0\}$), while Figure 1 reports the effect of self-consistency under different voting budgets ($n \in \{2, 4, 8, 16, 32\}$).

Table 5 shows that sampling temperature has a small and inconsistent effect on RVR. Qwen2.5-7B and Tülu-3-8B remain stable across temperatures on GSM8K and MATH, while Llama-3.1-

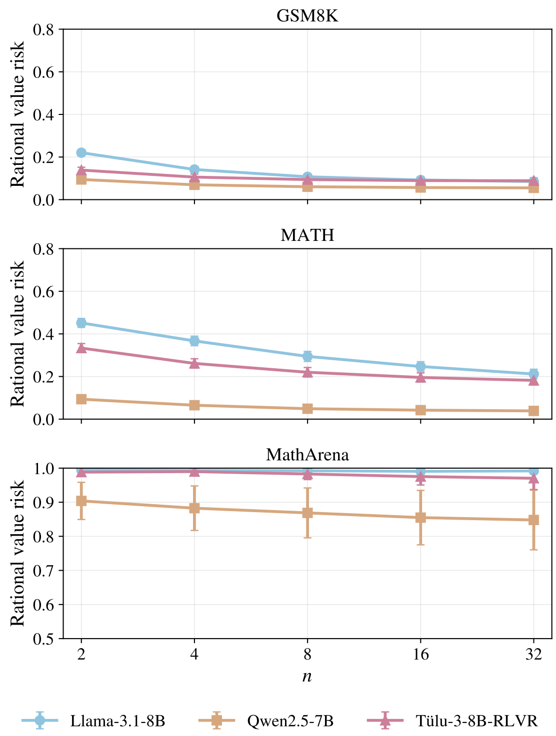

Figure 1: Effects of self-consistency under different voting budgets $n = \{2, 4, 8, 16, 32\}$ on rational value risk of Tülu-3-8B-RLVR, Qwen2.5-7B-Instruct (Qwen2.5-7B), and Llama-3.1-8B-Instruct (Llama-3.1-8B).

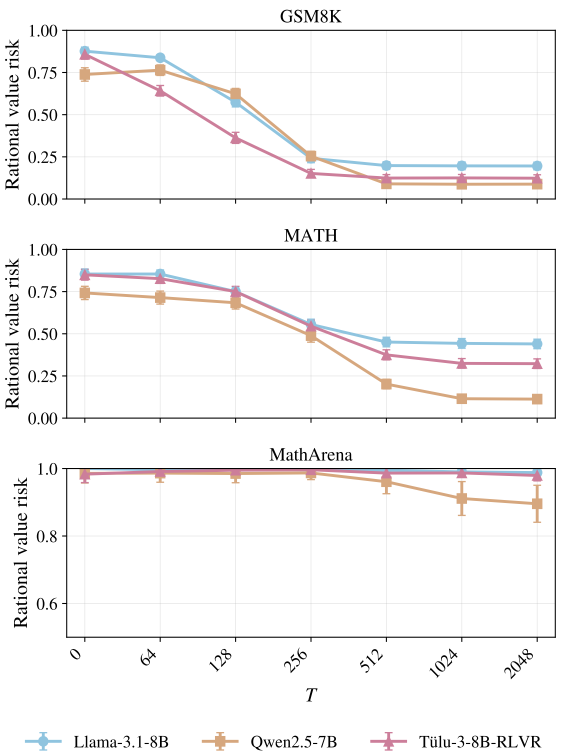

Figure 2: Rational value risk of Tülu-3-8B-RLVR, Qwen2.5-7B-Instruct (Qwen2.5-7B), and Llama-3.1-8B-Instruct (Llama-3.1-8B) vs. reasoning length T.

8B shows slightly larger increases at temperature $\tau = 1.0$. On MathArena, RVR stays consistently large across temperature settings. Figure 1 illustrates a different pattern for self-consistency. Increasing the voting budget reduces rational value risk, although the reduction becomes less apparent as n increases. On MATH, RVR decreases from 0.333 to 0.182 for Tülu-3-8B, from 0.094 to 0.039 for Qwen2.5-7B, and from 0.451 to 0.212 for Llama-3.1-8B as n increases from 2 to 32. Similar trends appear on GSM8K and MATH, but the improvement is weaker on MathArena.

These results indicate rational value risk depends strongly on self-consistency rather than sampling randomness. Temperature sampling improves the diversity of the sampled candidate set, but has a limited effect on rationality. In contrast, self-consistency with more voting budget contributes to rational reasoning.

H4: A longer chain of thought improves rationality with diminishing returns. Figure 2 illustrates the effect of reasoning length $T \in \{0, 64, 128, 256, 512, 1024, 2048\}$.

Rational value risk decreases with longer reasoning in most cases, although the rate of reduction decreases at longer chain of thought. On GSM8K, Tülu-3-8B-RLVR reduces RVR from 0.858 at $T = 0$ to 0.152 at $T = 256$, after which further reductions are slight, reaching 0.123 at $T = 2048$. Llama-3.1-8B also benefits from a larger reasoning budget, with RVR decreasing from 0.876 to 0.196. Qwen2.5-7B exhibits a small initial increase from 0.738 to 0.764, followed by a decrease to 0.088. A similar pattern appears on MATH, where for Tülu-3-8B-RLVR, RVR decreases steadily from 0.844 at $T = 0$ to 0.311 at $T = 2048$, with most of the reduction occurring before $T = 512$. These results show that longer reasoning can reduce rational value risk, but the additional benefit diminishes once the reasoning length is sufficiently large. In contrast, a longer chain of thought contributes less to improving rationality across all models on MathArena, where RVR remains high even with extended reasoning.

These results indicate that an appropriate reasoning-token budget can improve the rationality of LLM reasoning, while avoiding unnecessary token cost. However, simply extending the reasoning length is insufficient for harder deployment reasoning tasks.

6 Conclusions

This work identifies *rational value risk* as an inference-time failure that can persist after value alignment: the utility gap between a model's deployed reasoning strategy and its rational counterpart. We decompose the estimation error of this risk into bounded prompts, bounded responses, and imperfect verifiers, and evaluate it across open-weight and proprietary LLMs on conversational, mathematical, and code-generation benchmarks. Our results show that rational value risk is widespread, reduced but not eliminated by post-training, and can be partly mitigated by self-consistency and a longer chain of thought, suggesting rationality as a distinct evaluation dimension complementary to value alignment.

Limitations

Rationality over final answers vs. trajectories. In this work, we define rational value risk over final answers, which is the standard evaluation target in preference alignment, mathematical reasoning, and code generation. This allows rational value risk to be measured using existing outcome-based verifiers, such as preference judges, exact-match verifiers, unit tests, or symbolic checkers. We acknowledge that an alternative formulation could define this measure directly in terms of reasoning paths or intermediate reasoning steps. Process-level rationality evaluates the validity and efficiency of the reasoning trajectory toward the final answer, and may reveal failures that final-answer evaluation does not capture.

However, defining and verifying utility over reasoning processes introduces additional challenges, including comparing multiple reasoning paths, assessing partially correct intermediate steps, and evaluating latent or unfaithful reasoning traces. Therefore, we leave process-level rationality outside the scope of this work and focus on final-answer rationality as a broadly applicable and empirically measurable setting.

New algorithms for improved rationality. Another limitation of this work is that it does not propose a new inference or training algorithm for reducing rational value risk. Instead, our goal is to provide a formal definition, estimator, and empirical diagnosis of irrationality in LLM reasoning. This makes the framework primarily evaluative rather than prescriptive: it identifies when and where a model's reasoning strategy deviates from its rational counterpart, but does not directly provide a method for improvement.

Nevertheless, this work suggests several directions for future applications. Rational value risk can be utilised as an objective for designing inference-time algorithms, such as verifier-guided search, adaptive sampling, self-consistency, or compute allocation strategies that explicitly improve the expected utility of deployed reasoning. It may also inform post-training by distinguishing value misalignment from irrational reasoning under a well-aligned value function. In addition, this measure can be used as a diagnostic tool for comparing models, reasoning strategies, and deployment settings.

Human verification. Our evaluation of conversational tasks relies on LLM-based verifiers, without human annotations, which may limit the strength of the evaluation evidence. Since conversational tasks do not have ground-truth answers, human verification may provide stronger evidence for assessing verifier reliability. However, large-scale human evaluation is resource-intensive and difficult to scale, and is therefore out of the scope of this work.

Proprietary models. The experiments for H1 include proprietary models, which may limit reproducibility. In contrast, all models used for the experiments on verifiable reasoning tasks in H2-H4 are open-weight models, enabling reproducible evaluation in these settings.

Ethics considerations

This work is fundamental research. All experiments use publicly available datasets and benchmarks; no human subjects or sensitive data are involved. No direct negative societal impacts are identified.

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A Notation

Table 6: Notation

Symbol	Description
$\mathcal{X}$	Space of input prompts
$\mathcal{V}$	Finite vocabulary
$\mathcal{Z}$	Reasoning space, defined as $\mathcal{Z} = \bigcup_{T \geq 1} \mathcal{V}^T$
T	Length of a reasoning path
z	Reasoning path, $z = (z_1, \ldots, z_T)$
π_θ	Frozen language model parameterised by $\theta \in \Theta$
θ	Parameters of the language model
Θ	Space of language model parameters
$\mathcal{Y}$	Answer space
g	Extraction function, $g : \mathcal{Z} \to \mathcal{Y}$
D	Set of reasoning problems, $D = \{\mathbf{x}_i\}_{i=1}^M$
$\mathbf{x}_i$	Reasoning problem, $\mathbf{x}_i = (x_i, y_i^+)$
x_i	Input question
y_i^+	Preferred (or verifiable) answer
M	Number of reasoning problems
d_θ	Reasoning strategy, $d_\theta(\cdot \mid x_i) \triangleq d(\pi_\theta(\cdot \mid x_i))$
y_i	Answer extracted from the generated reasoning path, $y_i = g(z_i)$
$\mathcal{O}$	Outcome space
P	Outcome distribution of the verifier, $P : \mathcal{X} \times \mathcal{Y} \times \mathcal{Y} \to \Delta(\mathcal{O})$
o_i	Evaluation outcome, $o_i \sim P(\cdot \mid \mathbf{x}_i, y_i)$
f	Deterministic verifier, $f : \mathcal{X} \times \mathcal{Y} \times \mathcal{Y} \to \mathcal{O}$
U	Utility function defined on the outcome space, $U : \mathcal{O} \to \mathbb{R}$ with $U(o_i) \in [0, 1]$
K	Number of sampled reasoning responses for each reasoning problem
L	Number of verifier evaluations for each sampled response
$\mathcal{R}(d_\theta)$	Rational value risk of reasoning strategy d_θ
$\mathcal{R}_M(d_\theta)$	Rational value risk based on M sampled reasoning problems
$\overline{\mathcal{R}}_{M,K}(d_\theta)$	Rational value risk based on M sampled reasoning problems and K sampled reasoning responses for each problem
$\widehat{\mathcal{R}}_{M,K,L}(d_\theta)$	Estimated rational value risk based on M reasoning problems, K sampled reasoning responses, and L verifier evaluations

B Proofs

This appendix collects all proofs omitted from the main text.

B.1 Additional definitions

For notational brevity, we first define the expected utility of an answer y to a reasoning problem $\mathbf{x}$ as:

$$V(\mathbf{x}, y) \triangleq \mathbb{E}_{o \sim P(\cdot|\mathbf{x},y)}[U(o)].$$

Throughout this section, we assume $U(o) \in [0, 1]$, and therefore $V(\mathbf{x}, y) \in [0, 1]$.

For a given prompt $\mathbf{x}$, we define the expected utility of rational reasoning strategy d_{θ°,

$$V^\circ(\mathbf{x}) \triangleq \mathbb{E}_{y^\circ \sim d_{\theta^\circ}(\cdot|x)} V(\mathbf{x}, y^\circ),$$

and that of the deployed reasoning strategy d_θ,

$$V_\theta(\mathbf{x}) \triangleq \mathbb{E}_{y \sim d_\theta(\cdot|x)} V(\mathbf{x}, y).$$

Then, the rational value risk $\mathcal{R}(d_\theta)$ can be written as

$$\mathcal{R}(d_\theta) = \mathbb{E}_{\mathbf{x} \sim \rho}\left[V^\circ(\mathbf{x}) - V_\theta(\mathbf{x})\right].$$

Given M sampled prompts, we define

$$\mathcal{R}_M(d_\theta) = \frac{1}{M} \sum_{i=1}^{M} \left[V^\circ(\mathbf{x}_i) - V_\theta(\mathbf{x}_i)\right].$$

Given K sampled answers from the deployed reasoning strategy and its rational counterpart for every prompt, we define

$$\overline{\mathcal{R}}_{M,K}(d_\theta) = \frac{1}{MK} \sum_{i=1}^{M} \sum_{k=1}^{K} \left[V(\mathbf{x}_i, y^\circ_{i,k}) - V(\mathbf{x}_i, y_{i,k})\right],$$

where $y^\circ_{i,k} \overset{\text{iid}}{\sim} d_{\theta^\circ}(\cdot \mid x_i), \quad y_{i,k} \overset{\text{iid}}{\sim} d_\theta(\cdot \mid x_i)$.

Based on the definition 4, the estimated rational value risk is

$$\widehat{\mathcal{R}}_{M,K,L}(d_\theta) = \frac{1}{MK} \sum_{i=1}^{M} \sum_{k=1}^{K} \left[\widehat{U}_L(\mathbf{x}_i, y^\circ_{i,k}) - \widehat{U}_L(\mathbf{x}_i, y_{i,k})\right].$$

B.2 Proof of Lemma 1

Proof. Adding and subtracting $\mathcal{R}_M(d_\theta)$ and $\overline{\mathcal{R}}_{M,K}(d_\theta)$ gives

$$\begin{aligned}\mathcal{R}(d_\theta) - \widehat{\mathcal{R}}_{M,K,L}(d_\theta) =& \mathcal{R}(d_\theta) - \mathcal{R}_M(d_\theta) + \mathcal{R}_M(d_\theta) - \overline{\mathcal{R}}_{M,K}(d_\theta) \\ &+ \overline{\mathcal{R}}_{M,K}(d_\theta) - \widehat{\mathcal{R}}_{M,K,L}(d_\theta).\end{aligned}$$

This proves the claim. □

B.3 Proof of Lemma 2

Proof. For each sampled prompt and its preferred answer $\mathbf{x}_i$, we define

$$G_i \triangleq V^\circ(\mathbf{x}_i) - V_\theta(\mathbf{x}_i),$$

which measures the difference between the expected utility of the rational reasoning strategy d_{θ° and that of the deployed reasoning strategy d_θ.

Since $V^\circ(\mathbf{x}_i), V_\theta(\mathbf{x}_i) \in [0,1]$, their difference is bounded as $G_i \in [-1,1]$. Moreover, $G_1, \ldots, G_M$ are iid because $\mathbf{x}_1, \ldots, \mathbf{x}_M \overset{\text{iid}}{\sim} \rho$.

because the reasoning problems are sampled independently as $\mathbf{x}_1, \ldots, \mathbf{x}_M \overset{\text{iid}}{\sim} \rho$, the random variables $G_1, \ldots, G_M$ are independently and identically distributed as well.

By definition, we have

$$\mathbb{E}[G_i] = \mathcal{R}(d_\theta), \qquad \mathcal{R}_M(d_\theta) = \frac{1}{M}\sum_{i=1}^{M} G_i.$$

Based on the Hoeffding's inequality (Hoeffding, 1963), for any $\epsilon \in (0,1)$, their absolute difference can be bounded by

$$\Pr\left(|\mathcal{R}_M(d_\theta) - \mathcal{R}(d_\theta)| \geq \epsilon\right) \leq 2\exp\left(-\frac{M\epsilon^2}{2}\right). \tag{1}$$

For any $\delta \in (0,1)$, we set δ to the right-hand side of equation (1),

$$2\exp\left(-\frac{M\epsilon^2}{2}\right) = \delta.$$

Then, we have

$$\epsilon = \sqrt{\frac{2\log(2/\delta)}{M}}.$$

Therefore, with probability at least $1-\delta$,

$$|\mathcal{R}_M(d_\theta) - \mathcal{R}(d_\theta)| \leq \sqrt{\frac{2\log(2/\delta)}{M}},$$

which completes the proof.

□

B.4 Proof of Lemma 3

Proof. Given the sampled reasoning problems $\mathbf{x}_1, \ldots, \mathbf{x}_M$, we first define

$$Z_{i,k} \triangleq V(\mathbf{x}_i, y^\circ_{i,k}) - V(\mathbf{x}_i, y_{i,k}).$$

Since $V(\mathbf{x}, y) \in [0,1]$, their difference can be bounded as $Z_{i,k} \in [-1,1]$.

Then, the reasoning answers from rational reasoning strategy d_{θ° and deployed reasoning strategy d_θ are independent and we have

$$\mathbb{E}[Z_{i,k} \mid \mathbf{x}_i] = \mathbb{E}_{y^\circ \sim d_{\theta^\circ}(\cdot|x_i)} V(\mathbf{x}_i, y^\circ) - \mathbb{E}_{y \sim d_\theta(\cdot|x_i)} V(\mathbf{x}_i, y) = V^\circ(\mathbf{x}_i) - V_\theta(\mathbf{x}_i).$$

Consequently,

$$\frac{1}{MK}\sum_{i=1}^{M}\sum_{k=1}^{K} \mathbb{E}[Z_{i,k} \mid \mathbf{x}_i] = \mathcal{R}_M(d_\theta),$$

while

$$\frac{1}{MK}\sum_{i=1}^{M}\sum_{k=1}^{K} Z_{i,k} = \overline{\mathcal{R}}_{M,K}(d_\theta).$$

Applying Hoeffding's inequality (Hoeffding, 1963) conditionally on the M reasoning problems $\mathbf{x}_{1:M}$, we obtains

$$\Pr\left(\left|\overline{\mathcal{R}}_{M,K}(d_\theta) - \mathcal{R}_M(d_\theta)\right| \geq \epsilon \middle| \mathbf{x}_{1:M}\right) \leq 2\exp\left(-\frac{MK\epsilon^2}{2}\right). \tag{2}$$

Setting the right-hand side of equation (2) to δ, we have

$$\epsilon = \sqrt{\frac{2\log(2/\delta)}{MK}}.$$

With probability at least $1-\delta$,

$$\left|\overline{\mathcal{R}}_{M,K}(d_\theta) - \mathcal{R}_M(d_\theta)\right| \le \sqrt{\frac{2\log(2/\delta)}{MK}}$$

This proves the result. □

B.5 Proof of Lemma 4

Proof. For each sampled answer $y^\circ_{i,k}$ and $y_{i,k}$, let the outcome from rational reasoning be $o^\circ_{i,k,l} \overset{\text{iid}}{\sim} P(\cdot \mid \mathbf{x}_i, y^\circ_{i,k})$ and outcome from deployed reasoning be $o_{i,k,l} \overset{\text{iid}}{\sim} P(\cdot \mid \mathbf{x}_i, y_{i,k})$. We define

$$W_{i,k,l} \triangleq U(o^\circ_{i,k,l}) - U(o_{i,k,l}).$$

Since $U(o) \in [0,1]$, it can be bounded as $W_{i,k,l} \in [-1,1]$.

Given the sampled reasoning problems $\mathbf{x}_i$, we define the difference of expected utility between the rational and deployed reasoning answers, $y^\circ_{i,k}$, $y_{i,k}$, as

$$\mathbb{E}[W_{i,k,l} \mid \mathbf{x}_i, y^\circ_{i,k}, y_{i,k}] = V(\mathbf{x}_i, y^\circ_{i,k}) - V(\mathbf{x}_i, y_{i,k}).$$

Therefore, we have

$$\widehat{\mathcal{R}}_{M,K,L}(d_\theta) = \frac{1}{MKL}\sum_{i=1}^{M}\sum_{k=1}^{K}\sum_{l=1}^{L} W_{i,k,l},$$

and

$$\overline{\mathcal{R}}_{M,K}(d_\theta) = \frac{1}{MKL}\sum_{i=1}^{M}\sum_{k=1}^{K}\sum_{l=1}^{L} \mathbb{E}[W_{i,k,l} \mid \mathbf{x}_i, y^\circ_{i,k}, y_{i,k}].$$

Using Hoeffding's inequality (Hoeffding, 1963) conditionally on the reasoning problems $\mathbf{x}_i$ and reasoning answers, $y^\circ_{i,k}$ and $y_{i,k}$, we obtain

$$\Pr\left(\left|\widehat{\mathcal{R}}_{M,K,L}(d_\theta) - \overline{\mathcal{R}}_{M,K}(d_\theta)\right| \ge \epsilon\right) \le 2\exp\left(-\frac{MKL\epsilon^2}{2}\right). \tag{3}$$

Taking $\epsilon = \sqrt{2\log(2/\delta)/MKL}$, the upper bound on the left-hand side of equation (3) is δ. Therefore, with probability at least $1-\delta$,

$$\left|\widehat{\mathcal{R}}_{M,K,L}(d_\theta) - \overline{\mathcal{R}}_{M,K}(d_\theta)\right| \le \sqrt{\frac{2\log(2/\delta)}{MKL}}.$$

If the verifier is deterministic, for every reasoning problem $\mathbf{x}$ and its deployed reasoning answer y, we have $\widehat{U}_L(\mathbf{x}, y) = V(\mathbf{x}, y)$, and hence

$$\widehat{\mathcal{R}}_{M,K,L}(d_\theta) = \overline{\mathcal{R}}_{M,K}(d_\theta).$$

So the verification error is zero, which completes the proof. □

B.6 Proof of Theorem 1

Proof. By Lemma 1 and the triangle inequality,

$$\begin{aligned}\left|\mathcal{R}(d_\theta) - \widehat{\mathcal{R}}_{M,K,L}(d_\theta)\right| \leq |\mathcal{R}(d_\theta) - \mathcal{R}_M(d_\theta)| &+ \left|\mathcal{R}_M(d_\theta) - \overline{\mathcal{R}}_{M,K}(d_\theta)\right| \\ &+ \left|\overline{\mathcal{R}}_{M,K}(d_\theta) - \widehat{\mathcal{R}}_{M,K,L}(d_\theta)\right|.\end{aligned}$$

Apply Lemma 2, Lemma 3, and Lemma 4, each with failure probability $\delta/3$. By a union bound, these three events hold simultaneously with probability at least $1-\delta$. Combining the three bounds gives

$$\left|\mathcal{R}(d_\theta) - \widehat{\mathcal{R}}_{M,K,L}(d_\theta)\right| \leq \sqrt{\frac{2\log(6/\delta)}{M}} + \sqrt{\frac{2\log(6/\delta)}{MK}} + \sqrt{\frac{2\log(6/\delta)}{MKL}}.$$

This proves the result.

If the verifier is deterministic, Lemma 4 gives zero verification sampling error, and the third term vanishes. □

C Additional implementation details

This appendix provides additional details on the experiment configurations and implementations.

C.1 Sampling configuration

All experiments use a common sampling interface based on vLLM. Unless otherwise specified, we sample $K = 64$ candidate answers per prompt with

$$(\texttt{temperature}, \texttt{top\_p}, \texttt{top\_k}) = (1.0, 1.0, -1),$$

where $\texttt{top\_k} = -1$ disables top-k truncation. This default configuration is used for H1, H2, and H4. H3 varies the inference-time reasoning strategy by changing the sampling temperature and the self-consistency budget. For greedy decoding, we use $\tau = 0$ with $K = 1$, since multiple greedy samples would be identical.

Table 7 summarises the sampling configuration for each experiment.

Stop tokens. We do not configure additional stop tokens. For instruct models, the corresponding chat template already includes the assistant-end token, and generation stops when the EOS token is produced. For the Tülu-3 base model used in the few-shot H2 setting, no chat template is applied, and generation is controlled by `max_tokens`.

C.2 Prompting setup

Chat-mode prompts. For each chat-mode model and dataset pair, we apply the model's HuggingFace chat template to a fixed system–user message pair. The system prompt is dataset-specific and specifies the required answer format. The user prompt contains the input question or problem. This design keeps the prompting protocol consistent across models while allowing task-specific answer-format instructions.

```
GSM8K system: Solve the following math problem
step by step. After your reasoning, write the
final numeric answer on its own line in the
format: #### <answer>

MATH system: Solve the following math problem.
Show your reasoning, then put your final
answer in \boxed{}.

HumanEval system: Complete the following
Python function. Return ONLY the function
definition; do not include explanations or
examples.

MathArena system: Solve the following
competition math problem. Show your reasoning
step by step, then put your final answer
inside \boxed{}.

UltraFeedback system: (empty string)
AlpacaEval system: (empty string) – in
both, each generator/verifier applies its own
default system prompt.

User templates:
GSM8K : "question"
MATH : "problem"
MathArena : "problem"
HumanEval : "prompt"
UltraFeedback : "prompt"
AlpacaEval : "instruction"
```

Few-shot prompts for the Tülu-3 base model. The Tülu-3 base model does not use a chat template. In the H2 base-stage experiments, we prepare a fixed few-shot context for the user question. For GSM8K, we use the standard 8-shot chain-of-thought prompt from Wei et al. (2022). For maths, we use a 4-shot algebra prompt that matches the algebra-only split used in our experiments.

```
GSM8K 8-shot block (header lines only; full
text in repo):
Question: There are 15 trees in the grove.
...
Answer: ... The answer is 6.

Question: If there are 3 cars in the parking
lot ...
Answer: ... The answer is 5.
(... 6 more shots ...)

Question: Olivia has $23. She bought five
bagels for $3 each. ...
Answer: ... The answer is 8.

MATH 4-shot block (algebra-only, header lines
only):
Problem: Find the sum of all integer values
of n such that 20/(2n-1) is an integer.
Solution: ... The sum of these values is 1 +
0 + 3 + (-2) = \boxed{2}.

Problem: How many positive 3-digit integers
have all digits different?
Solution: ... 9 * 9 * 8 = \boxed{648}.

Problem: A right cylinder ... lateral surface
area?
Solution: ... 2 * pi * 2 * 5 = \boxed{20\pi}.

Problem: Compute sqrt((31)(30)(29)(28)+1).
Solution: ... = \boxed{869}.
```

Verifier prompts. For UltraFeedback, the verifier compares the generated response with the reference response and returns a win, a tie, or a lose. We use L verifier evaluations per candidate and return the averaged outcomes. These responses and pre-

Table 7: Sampling hyperparameters used in the experiments.

Experiment	τ	top-p	top-k	K	max_tokens	Notes
H1	1.0	1.0	-1	64	1024	7-8B models use max_tokens $= 512$ on AlpacaEval and UltraFeedback datasets.
H2	1.0	1.0	-1	64	1024	7-8B models use max_tokens $= 512$ on AlpacaEval and UltraFeedback datasets. Same decoding setting across SFT, DPO, and RLVR stages.
H3 direct, $\tau = 0$	0.0	1.0	-1	1	1024	Greedy decoding.
H3 direct, $\tau = 0.7$	0.7	1.0	-1	64	1024	Stochastic sampling.
H3 direct, $\tau = 1.0$	1.0	1.0	-1	64	1024	Default stochastic sampling.
H3 self-consistency	1.0	1.0	-1	64	1024	Voting budget $n \in \{2, 4, 8, 16, 32\}$.
H4	1.0	1.0	-1	64	T	$T \in \{0, 64, 128, 256, 512, 1024, 2048\}$.
Self-as-verifier calls	0.7	1.0	-1	1	8	$L = 5$ verifier calls per candidate.
External-verifier calls	0.7	1.0	-1	1	256	$L = 3$ verifier calls per candidate; DeepSeek-v4-flash.

Table 8: Self-as-verifier diagnostics on the UltraFeedback of H1 experiment. A-pick rate is the fraction of non-tie verifier calls in which the response in position A is selected. Krippendorff's α measures consistency among the $L = 5$ repeated verifier calls for the same candidate answer. It is computed as $1 - D_o/D_e$, where D_o is the observed disagreement among verifier calls and D_e is the average disagreement under random verifier evaluations. A larger value indicates more consistent verifier evaluations. A value close to 0.5 indicates small position bias.

Verifier	A-pick rate	Krippendorff's α	Mean |margin|
Tülu-3-8B-RLVR	0.502	0.599	3.63
Qwen2.5-7B-Instruct	0.501	0.595	3.67
Llama-3.1-8B-Instruct	0.502	0.639	3.82
Qwen2.5-72B-Instruct	0.441	0.732	3.59

ferred answer y^+ positions are randomised across calls to reduce position bias. Invalid evaluations are mapped to ties.

```
System: You are evaluating two responses to a
user's question. Pick the better one. Answer
with exactly one character: A if Response A
is better, B if Response B is better, T if
they are equally good.

User: User question: x
- Response A - a
- Response B - b

Which is better? Answer with one letter (A, B,
or T):
```

C.3 Benchmark details

GSM8K. We extract the final numeric answer using a priority order over common answer formats, including the canonical GSM8K answer marker, boxed answers, explicit "the answer is" statements, and the last numeric expression in the output. The predicted and reference answers are normalised before numerical comparison.

MATH and MathArena. We extract the predicted answer from the final expression `\boxed{}` and evaluate it using symbolic equivalence checking. In H1-H3, MATH answers that fail symbolic parsing are marked incorrect. To avoid confounding reasoning-length comparisons with parsing failures or missing boxed answers in H4. For MATH in H4, we prompt the model with `\n\nFinal answer: \boxed{` to encourage it to directly commit a final answer in `\boxed{}`. We then extract the final boxed answer and use a SymPy-based symbolic verifier to check whether it is mathematically equivalent to the reference answer. Outputs without a valid boxed answer are treated as unparseable. The MathArena benchmark in H1-H4 follows the same protocol.

HumanEval. Code generation tasks are evaluated by executing the generated solution against the provided unit tests in an isolated subprocess with time limits. A candidate receives utility 1 only if all tests pass, and utility 0 otherwise.

UltraFeedback. We use a self-as-verifier and an external verifier to compare each generated response with the reference answer y^+, respectively. The verifier returns win, tie, or lose, which is mapped to utilities 1, 0.5, and 0. We use multiple verifier evaluations and average their outcomes. The positions of generated and reference answers are randomised to reduce position bias. Table 8 reports diagnostics for the self-as-verifier setting and shows that the A/B position bias is small.

Table 9: Extraction functions and their conditional-correctness rates of the GSM8K

Model	#### N		\boxed{N}		"the answer is N"		Last number		No number		$M{\cdot}K$
	Fires	Acc.	Fires	Acc.	Fires	Acc.	Fires	Acc.	Fires	Acc.	
Tülu-3-8B-RLVR	99.9	86.2	< 0.1	62.5	—	—	< 0.1	25.0	—	—	84416
Qwen2.5-7B-Instruct	84.9	90.6	14.7	90.9	—	—	0.4	72.1	—	—	84416
Llama-3.1-8B-Instruct	1.6	74.9	< 0.1	78.3	22.0	83.4	76.3	76.5	< 0.1	—	84416

Table 10: MATH verify failure rates, reported as percentages of candidate answers.

Model	No \boxed{}	Empty GT	Parse error	Parse empty	Verify exception	Incorrect	Correct	$M{\cdot}K$
Tülu-3-8B-RLVR	4.9	—	—	< 0.1	—	29.3	65.8	64000
Qwen2.5-7B-Instruct	2.6	—	—	< 0.1	—	7.7	89.8	64000
Llama-3.1-8B-Instruct	27.7	—	—	< 0.1	—	25.2	47.0	64000

Table 11: GPU hours for each experiment.

Hypothesis	Number of cells	GPU-hours
H1	34	22.5
H2	20	7.7
H3	54	20.0
H4	68	26.4
Total	176	76.6

C.4 Answer extraction analysis

We further analyse answer-extraction failures for GSM8K and MATH. For GSM8K, Table 9 reports the used extraction function $g(\cdot)$ and the conditional correctness of each function. The residual unparseable rate is below 0.1%, indicating that the extractor covers almost all generated numeric answers.

For MATH, Table 10 shows that most failures come from missing final boxed answers, rather than symbolic parser errors. This means that the main source of $U = 0$ is the model's failure to provide the required final-answer format or a correct answer.

In the H4 experiment, we specify answer formats and extraction rules for each dataset. For GSM8K, we prompt the model with `\n\nFinal answer:` and extract the answer using the following formats in order:

- `#### N`;
- `\boxed{N}`;
- `the answer is N`;
- the last standalone number.

Table 12 reports the rate of unparseable outputs on GSM8K, MATH and MathArena across reasoning lengths T. The rates are nearly zero in most settings. One exception is Tülu-3-8B-RLVR on MathArena, which exhibits a relatively high unparseable rate for $T = 128$ to $T = 512$. However, this does not affect the conclusion in Figure 2, as rational value risk remains high on MathArena even at $T = 2048$.

C.5 Implementations

Experiments are run with Python 3.11, vLLM 0.6.3, PyTorch 2.5.1, CUDA 12.4, and Transformers 4.45.2. The main open-weight model experiments are run on six NVIDIA A800 GPUs with 80GB of memory. All reported confidence intervals are 95% prompt-bootstrap intervals with $B = 1000$ resamples. We use a fixed seed for each headline cell and cache generations by model, dataset, sampling configuration, prompt template, and sampling budget, so repeated runs reuse the same generated candidates when the configuration is unchanged.

C.6 Computational cost

The sampling stage requires 76.6 GPU-hours in total. Verification and bootstrap aggregation are performed separately and are not included in the GPU-hour count. Table 11 reports the GPU hours for each experiment.

D Effect of sample sizes on estimation error

We study the estimated rational value risk along the three terms in Theorem 1: the number of reasoning problems M, the number of sampled reasoning responses K, and the number of verifier verifications L.

Table 12: Fraction of unparseable answers across models on GSM8K, MATH, and MathArena with the increased reasoning lengths T in H4 experiment.

Dataset	Model	T=0	T=64	T=128	T=256	T=512	T=1024	T=2048
GSM8K	Tülu-3-8B-RLVR	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Qwen2.5-7B-Instruct	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Llama-3.1-8B-Instruct	0.18	0.02	0.00	0.00	0.00	0.00	0.00
MATH	Tülu-3-8B-RLVR	0.02	0.06	0.07	0.09	0.06	0.01	0.01
	Qwen2.5-7B-Instruct	0.00	0.06	0.02	0.02	0.03	0.00	0.00
	Llama-3.1-8B-Instruct	2.34	3.32	3.00	2.00	1.28	0.98	0.91
MathArena	Tülu-3-8B-RLVR	0.00	0.03	42.68	59.14	48.72	10.68	0.49
	Qwen2.5-7B-Instruct	0.00	0.00	0.00	0.00	0.03	0.00	0.00
	Llama-3.1-8B-Instruct	0.00	1.48	1.88	1.51	1.30	0.94	0.55

Table 13: Effect of compute budget K on rational value risk. Each entry is the prompt mean at budget K. The last column of each risk row includes the 95% prompt-bootstrap confidence interval.

Dataset	Model	K=1	K=2	K=4	K=8	K=16	K=32	K=64
GSM8K	Tülu-3-8B-RLVR	0.142	0.139	0.139	0.139	0.137	0.139	0.139
	Qwen2.5-7B-Instruct	0.093	0.096	0.093	0.093	0.094	0.093	0.094
	Llama-3.1-8B-Instruct	0.219	0.223	0.218	0.218	0.219	0.220	0.220
MATH	Tülu-3-8B-RLVR	0.318	0.331	0.336	0.338	0.339	0.340	0.342
	Qwen2.5-7B-Instruct	0.096	0.099	0.097	0.100	0.102	0.103	0.102
	Llama-3.1-8B-Instruct	0.542	0.526	0.528	0.528	0.528	0.531	0.530
HumanEval	Tülu-3-8B-RLVR	0.537	0.555	0.602	0.645	0.638	0.619	0.602
	Qwen2.5-7B-Instruct	0.238	0.238	0.239	0.242	0.250	0.245	0.251
	Llama-3.1-8B-Instruct	0.524	0.552	0.546	0.557	0.563	0.569	0.569
MathArena	Tülu-3-8B-RLVR	1.000	1.000	0.992	0.992	0.994	0.992	0.992
	Qwen2.5-7B-Instruct	0.900	0.900	0.912	0.915	0.914	0.915	0.913
	Llama-3.1-8B-Instruct	1.000	1.000	1.000	0.998	0.998	0.997	0.997

Table 14: Bootstrap CI width under different prompt sampling numbers M.

M	Tülu-3-8B	Qwen2.5-7B	Llama-3.1-8B
50	0.197±0.082	0.088±0.060	0.266±0.082
100	0.151±0.058	0.092±0.043	0.223±0.057
200	0.143±0.039	0.092±0.028	0.213±0.038
500	0.125±0.022	0.087±0.018	0.210±0.023
1319	0.139±0.014	0.094±0.012	0.220±0.013

Table 15: The number of reasoning problem M per experiment and dataset. Every cell draws K=64 samples per prompt; '-' marks a dataset not run in that experiment.

Dataset	H1	H2	H3	H4
AlpacaEval	300	300	300	-
UltraFeedback	300	300	500	-
GSM8K	1319	1319	1319	500
MATH	1000	1000	1000	500
MathArena	60	60	60	60
HumanEval	164	164	164	-

D.1 Effect of reasoning problem size

We assess prompt sampling error by bootstrapping subsets of prompts of size M. Table 14 shows that the confidence interval decreases as M grows, which is consistent with the $1/\sqrt{M}$ term in Theorem 1. Table 15 reports the number of reasoning problems M for each experiment and dataset.

D.2 Effect of reasoning response size

To assess the effect of the reasoning response size, we compute the estimator at truncated budgets $K \in \{1, 2, 4, 8, 16, 32, 64\}$ using the same cached candidate set. For each K, we compute the estimated rational value risk $\widehat{\mathcal{R}}_{K,M,L}$ in Definition 4,

$$\frac{1}{MK}\sum_{i=1}^{M}\sum_{k=1}^{K}\left[\widehat{U}_L(\mathbf{x}_i, y^{\circ}_{i,k}) - \widehat{U}_L(\mathbf{x}_i, y_{i,k})\right].$$

Table 13 reports the effect of reasoning response size on rational value risk in verifiable reasoning tasks with $L = 1$. The estimates are stable because M is sufficiently large, with only minor changes as K increases. In particular, the results remain nearly unchanged from $K = 32$ to $K = 64$ across the reported settings, suggesting that $K = 64$ provides a practical compute budget for estimating rational value risk.

Table 16: Rational value risk for each difficulty level at $K = 64$.

Dataset	Bucket	n	Tülu-3-8B-RLVR	Qwen2.5-7B-Instruct	Llama-3.1-8B-Instruct
MATH	L1	113	0.066	0.009	0.211
	L2	170	0.174	0.039	0.350
	L3	226	0.271	0.049	0.462
	L4	234	0.350	0.088	0.591
	L5	257	0.631	0.245	0.792
HumanEval	Short	56	0.571	0.165	0.462
	Medium	53	0.557	0.250	0.575
	Long	55	0.677	0.340	0.672

Table 17: Effect of L on rational value risk at $K = 32$ for the H1 experiment: Tülu-3-RLVR on UltraFeedback. (Std. is the standard deviation.)

L	DEU	Std.
1	0.558	0.0102
2	0.559	0.0094
3	0.560	0.0095
4	0.559	0.0098
5	0.559	0.0094

D.3 Effect of verifier evaluation size

For deterministic verifiers, such as GSM8K, MATH, HumanEval, and MathArena, the verifier outcome is fixed for a given answer. So repeated verifier verifications are not required, i.e., $L = 1$.

In conversational benchmarks, the number of verifier verifications L may affect the estimation error of a stochastic verifier. For UltraFeedback, we average the verifiers' outcomes at different L. Table 17 shows that the rational value risk estimation is stable across L, indicating that the verifier evaluation term is not the dominant source of estimation error in this setting, supported by Lemma 4.

E Rational value risk across various task difficulties

We provide a breakdown of difficulty for the MATH and HumanEval benchmarks to investigate the pattern of rational value risk across varying difficulty levels in Table Table 16.

For MATH, we use the official difficulty levels from 1 to 5. For HumanEval, which has no official difficulty label, we group problems by the token length of the reference solution into short, medium, and long groups.

The results show that rational value risk appears across difficulty levels. On MATH, the risk generally increases with difficulty level, especially for weaker models. On HumanEval, the risk also tends to increase with solution length.

F Statements on AI Assistants in Research and Writing

We used LLM assistants for language polishing, theory refinement, and code refactoring during the development of this work. All scientific claims, theoretical results, experimental results, and analyses were produced and verified by the authors; no AI-generated text was included without author review.